

MySQL

Installation & Configuration

Complete Tutorial — From Installation to Ready-to-Work

This comprehensive tutorial guides you through every step of setting up MySQL on Windows — from downloading the installer to securing your server, creating databases, managing users, and executing your first SQL queries. By the end of this tutorial, you will have a fully operational MySQL environment ready for development, training, or production use.

What You Will Learn

- | | |
|--|------------------------------------|
| ✓ Download & Install MySQL 8.0 on Windows | ✓ Secure your MySQL installation |
| ✓ Configure MySQL Server & set root password | ✓ Create databases, tables & users |
| ✓ Install & use MySQL Workbench (GUI) | ✓ Set user privileges & roles |
| ✓ Use MySQL Command Line Client (CLI) | ✓ Run your first SQL queries |

System Requirements

Component	Minimum	Recommended
Operating System	Windows 10 (64-bit)	Windows 10 / 11 (64-bit)
RAM	2 GB	4 GB or more
Disk Space	2 GB free	10 GB free
CPU	1 GHz single-core	Multi-core 2 GHz+
Microsoft Visual C++	2019 Redistributable	Latest version
Internet Connection	Required (download)	Broadband recommended

SECTION 1 — Downloading MySQL Installer

STEP 01

Navigate to the Official MySQL Download Page

Always download MySQL from the official source to ensure authenticity and security.

Download URL

```
https://dev.mysql.com/downloads/installer/
```

STEP 02

Choose the Correct Installer

On the download page, you will see two installer options:

Installer Type	File Size	When to Use
mysql-installer-web-community	~2 MB	Downloads components during setup (requires internet)
mysql-installer-community	~450 MB	Full offline installer — RECOMMENDED for training

✓ TIP

Select the larger full installer (mysql-installer-community). This avoids download interruptions during the setup wizard, especially important in training environments with slow internet.

STEP 03

Skip the Oracle Account Login

MySQL's download page may prompt you to create or log in to an Oracle account. You do not need to register.

- Scroll down below the login form
- Click the link: "No thanks, just start my download"
- The installer download will begin immediately

⚠ WARNING

Never download MySQL from third-party websites such as softonic.com or filehippo.com. These may bundle unwanted software or contain outdated, unsafe versions.

SECTION 2 — Running the MySQL Installer

STEP 04

Launch the Installer

1. Right-click the downloaded .msi file
2. Select "Run as Administrator" to ensure proper permission handling
3. If prompted by Windows User Account Control (UAC), click Yes
4. The MySQL Installer splash screen will load

⚠ WARNING

Always run the installer as Administrator. Running without elevated privileges can cause silent failures where MySQL installs but the service cannot start.

STEP 05

Choose a Setup Type

The installer presents five setup types. Select based on your intended use:

Setup Type	What It Installs	Best For
Developer Default	MySQL Server + Workbench + Shell + Connectors	Developers & learners — RECOMMENDED
Server Only	MySQL Server only (no GUI tools)	Dedicated server machines
Client Only	Client programs only (no server)	Remote database connections
Full	All MySQL products and tools	Complete installations
Custom	You choose individual components	Advanced / specific requirements

✓ TIP

For SQL training, data analytics, and general use — select Developer Default. This gives you MySQL Server 8.0 + MySQL Workbench (GUI) + MySQL Shell in one go.

STEP 06

Check Requirements

The installer will check for prerequisites such as Microsoft Visual C++ Redistributable. If any are missing:

- Click Execute to auto-install missing prerequisites
- Wait for all prerequisites to show a green checkmark
- Click Next to proceed

i NOTE

If Microsoft Visual C++ 2019 Redistributable is missing and the auto-install fails, download it manually from: https://aka.ms/vs/17/release/vc_redist.x64.exe

**STEP
07****Download & Install Products**

The installer will display a list of products to download and install. Review the list, then:

5. Click Execute — the installer downloads all selected components
6. A progress bar shows download and installation status for each product
7. Wait until all items show Status: Complete
8. Click Next to continue to configuration

SECTION 3 — Configuring the MySQL Server

After installation, the MySQL Installer automatically launches the Configuration Wizard. Follow each step carefully.

STEP 08 Type and Networking

Setting	Recommended Value	Notes
Config Type	Development Computer	Lower memory footprint; ideal for personal/training use
Connectivity	TCP/IP Enabled	Allows client connections
Port	3306	Default MySQL port — leave unchanged
X Protocol Port	33060	Used by MySQL Shell — leave unchanged
Open Firewall Port	Checked	Allows connections through Windows Firewall
Named Pipe / Shared Memory	Unchecked	Not needed for standard use

NOTE

Port 3306 is the standard MySQL port. Do not change it unless you have a specific reason (e.g., running multiple MySQL instances). Changing it requires updating all client connection strings.

STEP 09 Authentication Method

MySQL 8.0 introduced a stronger authentication plugin. Choose based on your environment:

Option	When to Choose
Use Strong Password Encryption (Recommended)	New projects, MySQL Workbench 8.0+, modern applications
Use Legacy Authentication Method	Only if connecting old applications that don't support SHA-256

WARNING

Always select Strong Password Encryption (`caching_sha2_password`) for new installations. Legacy mode is a security downgrade and should only be used for backward compatibility with old software.

STEP 10 Set the Root Password

The root account is the MySQL superuser — it has full access to everything in the database server.

Root Password Requirements

- Minimum 8 characters

- Mix of uppercase and lowercase letters
- Include at least one number
- Include at least one special character (e.g., @, #, !, %)

⚠ WARNING

CRITICAL: Write down your root password and store it securely. If you lose the root password, recovering access to MySQL requires stopping the service, starting it in skip-grant-tables mode, and manually resetting — a complex process that risks data exposure.

Adding Additional User Accounts (Optional at This Stage)

The installer allows you to add extra MySQL user accounts now. You may skip this and add users later via SQL commands. Refer to Section 6 of this tutorial for user management.

STEP 11 Windows Service Configuration

MySQL runs as a Windows Service, which means it starts automatically with Windows and runs in the background.

Setting	Recommended	Purpose
Configure as Windows Service	Checked	Runs MySQL as a background service
Windows Service Name	MySQL80	Service identifier in Windows
Start at System Startup	Checked	Auto-starts MySQL when Windows boots
Run Windows Service as	Standard System Account	Default; use for development machines

✓ TIP

The service name MySQL80 is used to start/stop MySQL from the command line: net start MySQL80 or net stop MySQL80. Remember this for troubleshooting.

STEP 12 Apply Configuration

Review the configuration summary, then click Execute. The installer will:

9. Write the MySQL configuration file (my.ini)
10. Update the Windows Registry with MySQL settings
11. Create the MySQL data directory (usually C:\ProgramData\MySQL\MySQL Server 8.0\Data)
12. Start the MySQL80 Windows Service
13. Apply security settings

⚠ WARNING

If any step shows an error (red X), do not proceed. Take a screenshot of the error, note the step number, and check the MySQL error log at: C:\ProgramData\MySQL\MySQL Server 8.0\Data\<hostname>.err

SECTION 4 — Verifying the Installation

STEP 13

Verify MySQL Service is Running

Method A: Using Windows Services

14. Press Windows + R, type services.msc, press Enter
15. Scroll to find MySQL80 in the list
16. Status should show Running

Method B: Using Command Prompt

```
-- Open Command Prompt as Administrator, then run:
sc query MySQL80

-- Expected output:
SERVICE_NAME: MySQL80
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 4   RUNNING
```

Method C: Using Task Manager

17. Press Ctrl + Shift + Esc to open Task Manager
18. Go to the Services tab
19. Find MySQL80 — Status should be Running

STEP 14

Test Connection via MySQL Command Line Client

MySQL installs a dedicated command-line client. Let us use it to verify the server is accepting connections.

20. Open the Start Menu and search for "MySQL 8.0 Command Line Client"
21. Click to open — a terminal window will appear
22. Enter your root password when prompted
23. You should see the MySQL prompt:

```
Enter password: ****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.36 MySQL Community Server - GPL

mysql>
```

Run a Quick Verification Query

```
mysql> SELECT VERSION();

+-----+
```

```
| VERSION() |  
+-----+  
| 8.0.36 |  
+-----+  
1 row in set (0.00 sec)
```

✓ TIP

If you see the mysql> prompt and can run queries, your MySQL installation is successful and the server is ready.

**STEP
15****Add MySQL to System PATH (Environment Variables)**

Adding MySQL to the PATH lets you run mysql commands from any Command Prompt window without navigating to the installation folder.

24. Press Windows + S, search for "Environment Variables"
25. Click "Edit the system environment variables"
26. In the System Properties window, click "Environment Variables..."
27. Under System Variables, find and select Path, then click Edit
28. Click New and add the MySQL bin path:

```
C:\Program Files\MySQL\MySQL Server 8.0\bin
```

29. Click OK on all open windows
30. Open a NEW Command Prompt and test:

```
mysql --version  
  
-- Expected output:  
mysql Ver 8.0.36 Distrib 8.0.36, for Win64 (x86_64)
```


SECTION 5 — Using MySQL Workbench (GUI)

MySQL Workbench is the official graphical interface for MySQL. It provides a visual environment for database design, query writing, server administration, and data import/export.

STEP 16 Launch MySQL Workbench

31. Open the Start Menu and search "MySQL Workbench"
32. Click to launch — the Workbench home screen appears
33. Under MySQL Connections, you will see a tile labelled Local instance MySQL80
34. Click the tile to connect
35. Enter your root password when prompted
36. Tick "Save password in vault" to avoid re-entering it each session

NOTE

If no connection tile exists, click the + icon next to MySQL Connections to create one manually. Use hostname: 127.0.0.1, Port: 3306, Username: root.

STEP 17 Workbench Interface Overview

Panel / Area	Description
Navigator (Left)	Shows Schemas (databases), Administration, and Instance tabs
Query Editor (Centre)	Write and execute SQL queries here
Result Grid (Bottom)	Displays query output in a tabular format
Output / Log (Bottom)	Shows execution messages, errors, and timing
Toolbar (Top)	Quick access to run query, save, format SQL, etc.
Schema Panel (Left)	Right-click on any table to inspect, select rows, or edit data

STEP 18 Key Workbench Keyboard Shortcuts

Shortcut	Action
Ctrl + Enter	Execute the current query (cursor line or selection)
Ctrl + Shift + Enter	Execute ALL queries in the editor
Ctrl + /	Comment / uncomment selected lines
Ctrl + B	Auto-format (beautify) SQL code
Ctrl + W	Close the current query tab
Ctrl + T	Open a new query tab
F5	Refresh the Schema Navigator
Ctrl + Z	Undo last action

Shortcut	Action
Ctrl + S	Save the current script file

SECTION 6 — Securing Your MySQL Installation

A freshly installed MySQL server has several default settings that can be security risks in production and shared environments. Run the following steps to harden your installation.

STEP 19 Run `mysql_secure_installation`

This built-in utility removes insecure defaults. Open Command Prompt as Administrator and run:

```
mysql_secure_installation -u root -p
```

The utility will ask a series of questions. Use the recommended responses below:

Question	Recommended Response	Reason
Would you like to setup VALIDATE PASSWORD component?	Y	Enforces strong passwords
Password validation policy level (0, 1, 2)?	1 (MEDIUM)	Balance of security and usability
Remove anonymous users?	Y	Prevents unauthenticated access
Disallow root login remotely?	Y	Root should only log in locally
Remove test database and access to it?	Y	Test DB is publicly accessible by default
Reload privilege tables now?	Y	Applies all changes immediately

STEP 20 Change the Root Password (If Needed)

If you need to update the root password at any time, connect to MySQL and run:

```
ALTER USER 'root'@'localhost'  
IDENTIFIED BY 'YourNewStrongPassword@2024';  
  
FLUSH PRIVILEGES;
```

STEP 21 Disable Remote Root Login

Verify that root cannot log in from remote hosts (should already be set by `mysql_secure_installation`):

```
SELECT user, host FROM mysql.user WHERE user = 'root';  
  
-- You should only see host = 'localhost', not '%'  
-- If '%' exists, remove it:
```

```
DELETE FROM mysql.user WHERE user='root' AND host='%';  
FLUSH PRIVILEGES;
```

SECTION 7 — Creating & Managing MySQL Users

Best practice: Never use the root account for day-to-day database work. Create dedicated user accounts with the minimum privileges needed for each task.

STEP 22 Create a New User

Syntax

```
CREATE USER 'username'@'host'  
IDENTIFIED BY 'password';
```

Examples

```
-- Create a user that can only connect from localhost  
CREATE USER 'analyst'@'localhost'  
IDENTIFIED BY 'Analyst@Secure2024';  
  
-- Create a user that can connect from any host (use cautiously)  
CREATE USER 'app_user'@'%'  
IDENTIFIED BY 'AppUser@Secure2024';  
  
-- Create a user for a specific IP address  
CREATE USER 'reports'@'192.168.1.100'  
IDENTIFIED BY 'Reports@Secure2024';
```

STEP 23 Grant Privileges

Common Privilege Levels

Privilege Level	Syntax Example	Grants Access To
All on all DBs	GRANT ALL PRIVILEGES ON *.* TO 'user'@'host'	Everything (admin equivalent)
All on one DB	GRANT ALL ON mydb.* TO 'user'@'host'	Full access to one database
Read only	GRANT SELECT ON mydb.* TO 'user'@'host'	View data only — cannot modify
Read + Write	GRANT SELECT, INSERT, UPDATE ON mydb.* TO 'user'@'host'	View and modify data
Specific table	GRANT SELECT ON mydb.sales TO 'user'@'host'	Single table access only

Practical Examples

```
-- Grant full access to 'analyst' user on the 'company_db' database  
GRANT ALL PRIVILEGES ON company_db.*
```

```
TO 'analyst'@'localhost';

-- Grant read-only access to a reporting user
GRANT SELECT ON company_db.*
TO 'reports'@'192.168.1.100';

-- Apply the privilege changes immediately
FLUSH PRIVILEGES;
```

**STEP
24****View, Revoke & Drop Users**

```
-- View all users in the system
SELECT user, host FROM mysql.user;

-- View privileges for a specific user
SHOW GRANTS FOR 'analyst'@'localhost';

-- Revoke a specific privilege
REVOKE INSERT ON company_db.*
FROM 'analyst'@'localhost';

-- Remove all privileges
REVOKE ALL PRIVILEGES ON company_db.*
FROM 'analyst'@'localhost';

-- Delete the user account completely
DROP USER 'analyst'@'localhost';

FLUSH PRIVILEGES;
```

SECTION 8 — Creating Databases and Tables

STEP 25

Create a New Database

```
-- Create a new database
CREATE DATABASE company_db;

-- Create with explicit character set (recommended for multilingual data)
CREATE DATABASE company_db
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

-- View all databases
SHOW DATABASES;

-- Switch to (use) the database
USE company_db;
```

✓ TIP

Always use CHARACTER SET utf8mb4 for new databases. It supports all Unicode characters including Malay, Arabic, Chinese, and Emoji. The older utf8 in MySQL is limited to 3 bytes and cannot store some characters.

STEP 26

Create Tables

A well-designed table includes data types, constraints, and a primary key. Study the example below:

```
USE company_db;

-- Create an employees table
CREATE TABLE employees (
  employee_id INT NOT NULL AUTO_INCREMENT,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) NOT NULL UNIQUE,
  department VARCHAR(50),
  hire_date DATE NOT NULL,
  monthly_salary DECIMAL(10,2) DEFAULT 0.00,
  is_active TINYINT(1) DEFAULT 1,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (employee_id)
);

-- Verify the structure
DESCRIBE employees;
```

MySQL Common Data Types Reference

Data Type	Example	Use For
INT	INT NOT NULL	Whole numbers, IDs, counts

Data Type	Example	Use For
DECIMAL(p,s)	DECIMAL(10,2)	Money, precise decimals (p=precision, s=scale)
VARCHAR(n)	VARCHAR(100)	Variable-length text up to n characters
TEXT	TEXT	Long text content (no length limit needed)
DATE	DATE	Date only: YYYY-MM-DD
DATETIME	DATETIME	Date and time: YYYY-MM-DD HH:MM:SS
TIMESTAMP	TIMESTAMP	Auto-records current date/time on INSERT
TINYINT(1)	TINYINT(1) DEFAULT 1	Boolean-like flag: 0 = false, 1 = true
ENUM	ENUM('Active','Inactive')	Fixed set of allowed values

**STEP
27**
Insert, View & Modify Data

```
-- Insert a single record
INSERT INTO employees (full_name, email, department, hire_date, monthly_salary)
VALUES ('Ahmad Fauzi', 'ahmad@company.com', 'IT', '2022-03-15', 4500.00);

-- Insert multiple records at once
INSERT INTO employees (full_name, email, department, hire_date, monthly_salary)
VALUES
    ('Siti Rahmah', 'siti@company.com', 'HR', '2021-07-01', 4200.00),
    ('Raj Kumar', 'raj@company.com', 'Finance', '2023-01-10', 5100.00),
    ('Lin Wei', 'lin@company.com', 'IT', '2020-05-22', 5800.00);

-- View all records
SELECT * FROM employees;

-- Update a record
UPDATE employees
SET monthly_salary = 4800.00
WHERE employee_id = 1;

-- Delete a record
DELETE FROM employees
WHERE employee_id = 4;
```


SECTION 9 — Running Your First SQL Queries

Now that the server is configured and a sample table exists, practice these essential queries to confirm everything is working correctly.

Essential Query Patterns

```
-- 1. View all data
SELECT * FROM employees;

-- 2. Select specific columns only
SELECT full_name, department, monthly_salary
FROM employees;

-- 3. Filter with WHERE
SELECT * FROM employees
WHERE department = 'IT';

-- 4. Sort results
SELECT full_name, monthly_salary
FROM employees
ORDER BY monthly_salary DESC;

-- 5. Aggregate: count employees per department
SELECT department, COUNT(*) AS headcount
FROM employees
GROUP BY department
ORDER BY headcount DESC;

-- 6. Average salary per department
SELECT department,
       ROUND(AVG(monthly_salary), 2) AS avg_salary
FROM employees
GROUP BY department;

-- 7. Find employees hired in 2022 or later
SELECT full_name, hire_date
FROM employees
WHERE hire_date >= '2022-01-01'
ORDER BY hire_date ASC;

-- 8. Search by partial name (LIKE)
SELECT * FROM employees
WHERE full_name LIKE 'Ahmad%';
```

SECTION 10 — Backup & Restore

Regular backups are critical. MySQL provides mysqldump for creating SQL-based backups that can be restored on any MySQL instance.

Export (Backup) a Database

```
-- Open Command Prompt (with MySQL in PATH), then run:
-- Syntax: mysqldump -u [user] -p [database_name] > [output_file.sql]

mysqldump -u root -p company_db > C:\Backup\company_db_backup.sql

-- Backup with timestamp in filename (useful for scheduling)
-- Run this in PowerShell:
mysqldump -u root -p company_db > "C:\Backup\company_db_$(Get-Date -Format
yyyyMMdd_HHmm).sql"
```

Import (Restore) a Database

```
-- Method 1: From Command Prompt
mysql -u root -p company_db < C:\Backup\company_db_backup.sql

-- Method 2: From inside MySQL prompt
mysql> USE company_db;
mysql> SOURCE C:/Backup/company_db_backup.sql;

-- Method 3: Via MySQL Workbench
-- Server menu > Data Import > Import from Self-Contained File
-- Select the .sql file > Select Target Schema > Start Import
```

⚠ WARNING

Use forward slashes (/) or double backslashes (\\) in file paths when inside the MySQL prompt. Windows-style single backslashes (\) are interpreted as escape characters and will cause errors.

SECTION 11 — Troubleshooting Common Issues

Error / Issue	Likely Cause	Solution
Can't connect to MySQL server on '127.0.0.1' (10061)	MySQL service is not running	Open Services, find MySQL80, click Start
Access denied for user 'root'@'localhost'	Wrong password	Reset password via safe mode (see below)
mysql is not recognized as an internal command	MySQL not in system PATH	Add C:\Program Files\MySQL\MySQL Server 8.0\bin to PATH
ERROR 1045 (28000)	Authentication failure	Verify username, host, and password are correct
Port 3306 already in use	Another app uses the port	Stop the conflicting app, or change MySQL port in my.ini
MySQL Installer hangs at 'Starting service'	Windows Firewall blocking service	Temporarily disable firewall and retry; then add exception
ERROR 1820: You must reset your password	Temporary password expired	Run: ALTER USER 'root'@'localhost' IDENTIFIED BY 'NewPass@1';
Table 'db.table' doesn't exist	Wrong database selected	Run: USE database_name; before your query
ERROR 2006: MySQL server has gone away	Connection timeout or packet size too large	Increase max_allowed_packet in my.ini

Reset Root Password (Emergency Procedure)

```
-- Step 1: Stop the MySQL service
net stop MySQL80

-- Step 2: Start MySQL without authentication (safe mode)
mysqld --skip-grant-tables --shared-memory

-- Step 3: Open a new Command Prompt and connect
mysql -u root

-- Step 4: Reset the password
FLUSH PRIVILEGES;
ALTER USER 'root'@'localhost' IDENTIFIED BY 'NewPassword@2024';
FLUSH PRIVILEGES;
EXIT;

-- Step 5: Stop the safe mode instance (Ctrl+C in its window)
-- Step 6: Restart MySQL normally
net start MySQL80
```

Key MySQL Configuration File Locations

Item	Default Location
Configuration file	C:\ProgramData\MySQL\MySQL Server 8.0\my.ini
Data directory	C:\ProgramData\MySQL\MySQL Server 8.0\Data\

Item	Default Location
Error log	C:\ProgramData\MySQL\MySQL Server 8.0\Data\<hostname>.err
Binlog directory	C:\ProgramData\MySQL\MySQL Server 8.0\Data\
MySQL executables	C:\Program Files\MySQL\MySQL Server 8.0\bin\

SECTION 12 — Essential Admin Commands Reference

```
-- — SERVER STATUS & INFO —————
SELECT VERSION();           -- MySQL version
SELECT NOW();               -- Current server date & time
SHOW STATUS;               -- Server status variables
SHOW VARIABLES LIKE 'max_connections'; -- Check a specific variable
SHOW PROCESSLIST;          -- Active connections/queries

-- — DATABASE MANAGEMENT —————
SHOW DATABASES;             -- List all databases
CREATE DATABASE mydb CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE mydb;                   -- Switch to a database
DROP DATABASE mydb;         -- Delete a database (IRREVERSIBLE)
SELECT DATABASE();          -- Show currently selected database

-- — TABLE MANAGEMENT —————
SHOW TABLES;              -- List tables in current database
DESCRIBE tablename;         -- Show table column structure
SHOW CREATE TABLE tablename; -- Show full CREATE statement
RENAME TABLE old_name TO new_name; -- Rename a table
DROP TABLE tablename;      -- Delete a table (IRREVERSIBLE)

-- — USER MANAGEMENT —————
SELECT user, host FROM mysql.user; -- List all user accounts
SHOW GRANTS FOR 'user'@'host';     -- Show user privileges
FLUSH PRIVILEGES;                  -- Reload privilege tables

-- — PERFORMANCE —————
EXPLAIN SELECT * FROM tablename WHERE id = 1; -- Query execution plan
SHOW INDEX FROM tablename;                   -- List indexes on a table
OPTIMIZE TABLE tablename;                   -- Reclaim space and defragment
```

SETUP COMPLETION CHECKLIST

Done	Checklist Item
☐	MySQL 8.0 downloaded from official source (dev.mysql.com)
☐	Installer run as Administrator
☐	Developer Default setup type selected
☐	Root password set and stored securely
☐	MySQL80 Windows service running
☐	Connection verified via MySQL Command Line Client
☐	MySQL Workbench opened and connected successfully
☐	MySQL added to system PATH environment variable
☐	mysql_secure_installation executed
☐	Anonymous users and test database removed
☐	Dedicated user account created (not using root for daily work)

Done	Checklist Item
☐	Test database and table created successfully
☐	First SELECT query executed and returned results
☐	Backup (mysqldump) tested and verified

You are now ready to work with MySQL. Proceed to the SQL Case Study exercises to build your querying skills.

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